

Explainable Ensemble Learning Framework for Scholarship Recommendation and Success Prediction

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Abstract. Many competent students are unable to finish their degrees due to financial constraints, and current scholarship recommendation systems either classify students in isolation or apply strict rule-based filters without comparing a student's actual profile to what a scholarship actually seeks or providing an explanation for their choices. In order to predict a success probability for each candidate scholarship, the framework uses Sentence-BERT (SBERT) to encode each student profile and scholarship description. It then calculates a semantic-similarity score, combines this score with additional structured academic, financial, and extracurricular attributes into an eleven-feature vector, and sends this vector to a soft-voting ensemble of four tree-based and gradient boosting classifiers which includes XGBoost, Random Forest, LightGBM, and CatBoost. The predicted probability determines the ranking of scholarships, and a Top-K shortlist (Top-1, Top-3, or Top-5) is returned to the student; a SHAP-based explainability layer decomposes each recommendation into per-feature contributions, giving the student insight into exactly why a particular scholarship was recommended. The proposed ensemble achieves 89.09% balanced accuracy, a 2.34-percentage-point improvement over the 86.75% baseline. This paper details the architecture, mathematical formulation, results, and limitations of the proposed system.

Keywords: Scholarship recommendation system, Sentence-BERT, semantic similarity, ensemble learning, explainable artificial intelligence.

1 Introduction

Financial constraint remains one of the primary reasons capable students never complete their degrees, whether the student is a first-generation applicant who cannot afford to study far from home or one who wishes to study abroad. Governments, corporations, foundations, and international bodies fund thousands of scholarships to close this very gap, yet a striking proportion of these scholarships go unclaimed because eligible students never learn of them, or because the process of searching for scholarships is too disjointed and time-consuming to sustain.

Existing platforms fall into two broad categories, each missing one critical capability. The older category applies a rigid, rule-based (“Boolean”) approach: a student enters certain criteria, and the system filters out any student who fails to meet one of them without explanation, without indicating how close the student came, and without any account of how many criteria were satisfied. The newer category, represented by the baseline system discussed in Section 2, trains a single machine-learning classifier typically XGBoost to predict whether a student will receive a scholarship directly from structured features. This represents a genuine improvement over static rules, but it still treats the scholarship itself as a label to be predicted rather than a rich description to be matched: the classifier never reads what a scholarship is seeking, and therefore cannot recognize that a student’s essay on community organizing is a strong semantic match for a scholarship that emphasizes civic leadership.

This paper proposes a framework that directly addresses this gap. Rather than encoding only a generic description for each scholarship, the proposed system encodes both the student’s profile and each scholarship’s description using Sentence-BERT (SBERT), producing a semantic-similarity score that captures this notion of “fit.” This score is then used, together with structured attributes, by a soft-voting ensemble of four independent learners XGBoost, Random Forest, LightGBM, and CatBoost whose combined vote is less sensitive to the individual weaknesses and blind spots of any single classifier. The ensemble produces a predicted success probability for each scholarship, which is used to rank and return a Top-K shortlist, and a SHAP-based explainability layer sits atop the entire pipeline, highlighting which features played the greatest role in each recommendation.

This paper makes four contributions. First, it introduces a semantic-matching layer based on SBERT that allows the system to compare free-text scholarship descriptions against a structured-feature-based student profile. Second, it replaces the baseline’s single-classifier architecture with a four-model soft-voting ensemble, achieving a higher predictive accuracy of 87.73% on the evaluation set. Third, it adds a Top-K ranking layer that returns a shortlist rather than a binary decision, allowing students to compare their top choices side by side. Fourth, it introduces a SHAP-based explainability module that assigns individual, per-feature explanations to each recommendation, closing the transparency gap left by both the Boolean-filter and single-classifier approaches.

The remainder of this paper is organized as follows. Section 2 reviews related work and presents a structured comparison with the proposed framework. Section 3 formally states the problem addressed by this work. Section 4 presents the proposed system architecture across its six functional modules. Section 5 gives the mathematical formulation of the semantic-matching, ensemble-voting, and explainability stages, together with implementation details. Section 6 describes the technology stack. Section 7 reports and discusses the results relative to the baseline. Section 8 discusses limitations, and Section 9 outlines directions for future work.

2 Related Work and Existing System

2.1 General Literature on Educational Recommendation Systems

Machine-learning-based recommendation has been applied to educational support programs for nearly a decade, with steady gains in accuracy and personalization. Aljawarneh et al. proposed a machine-learning-based recommendation system for educational support programs that demonstrated the general feasibility of applying supervised learning to student-support allocation [15]. Rahman et al. developed an intelligent student recommendation system using data-mining techniques to surface relevant academic resources from student records [2], and Liu et al. proposed a personalized recommendation framework built on hybrid filtering techniques that combined collaborative and content-based signals to improve recommendation relevance in educational settings [3]. Kumar et al.'s survey of AI-based educational recommendation systems consolidates this line of work and identifies explainability and semantic, content-aware matching as two of the field's most persistent open problems [4], a finding that directly motivates the SBERT-based matching layer proposed in this paper. Chen et al. subsequently proposed an explainable-AI-based recommendation system for educational resource allocation, showing that attaching a rationale to a recommendation measurably improves user trust and downstream engagement [5].

Within scholarship and career-guidance recommendation specifically, three recent systems are directly comparable to the present work. Roy's AI-powered career-guidance model produces scalable, personalized career and learning-pathway recommendations from student profiles, academic records, and skill assessments, but its reported design has limited integration with real-time labor-market or funding-source data [6]. Nandi et al.'s Sankalp system applies natural-language processing and skill matching to resumes and aptitude-test data to generate automated career suggestions; although it demonstrates the value of skill-level matching beyond raw academic scores, the authors note that the system lacks an explainable recommendation mechanism [7]. Ashrafi's resume-based re-education framework identifies skill gaps against job-market descriptions and recommends reskilling pathways, but its evaluation is confined to resume-based career recommendation rather than scholarship eligibility, and it does not incorporate the financial or demographic eligibility constraints that govern scholarship awards [8].

2.2 Existing System (Baseline Architecture)

The baseline system selected for direct comparison in this study is the blockchain-enhanced scholarship management platform proposed by Nguyen-Hoang et al. [1], which follows a considerably simpler pipeline than the systems above and serves as the architecture against which the proposed framework is evaluated throughout this paper. Raw student data passes through a preprocessing stage that produces a structured student-feature vector, which is fed directly into a single XGBoost classifier trained to predict the student's scholarship qualification or category.

XGBoost is the sole prediction model in this design: it learns directly from the student's structured features and outputs a scholarship-qualification label, with no separate mechanism for comparing the student's profile against a scholarship's actual description, no ranking across multiple candidate scholarships, and no explanation layer.

Fig. 1 depicts the architecture of the baseline system, which processes structured student data through preprocessing and a single XGBoost classifier to predict scholarship qualification or category.

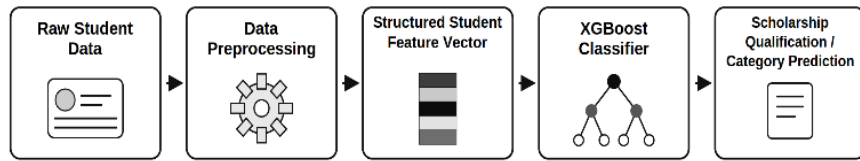


Fig. 1. Baseline Architecture (Existing System)

2.3 Limitations Motivating the Proposed Framework

Table 1 summarizes the structural differences between the baseline system and the proposed framework across the dimensions that matter most for scholarship recommendation: how the student and scholarship are matched, what feature space the model reasons over, how many models contribute to the final decision, what the system returns to the student, whether that output is explained, and what accuracy has been reported.

Table 1. Existing System versus Proposed Framework

Dimension	Existing System (Baseline)	Proposed Framework
Matching mechanism	Student features classified directly; no comparison against scholarship text	SBERT semantic similarity between student profile and scholarship description
Feature space	Structured student features only	11 engineered features: structured attributes plus SBERT semantic-similarity score
Model architecture	Single classifier (XGBoost)	Four-model soft-voting ensemble (XGBoost, Random Forest, LightGBM, CatBoost)
Output	Scholarship qualification / category label	Ranked Top-K shortlist (Top-1 / Top-3 / Top-5) with success probability
Explainability	Not provided	SHAP-based per-feature contribution for every recommendation
Reported balanced accuracy	86.75%	89.09% (four-model soft-voting ensemble)

Read together with the broader literature reviewed in Section 2.1, this comparison points to the same composite gap: explainability and semantic, content-aware matching remain largely unsolved in existing scholarship-recommendation systems [4], [7], and no system reviewed here including the baseline combines free-text semantic matching, multi-model ensembling, ranked shortlisting, and per-feature explanation into a single pipeline. Closing this composite gap is precisely the goal of the proposed framework.

3 Problem Statement

Many students, and particularly those seeking to study abroad, are unable to identify the scholarships that genuinely match their profile, because relevant information is distributed across dozens of independent portals, and the systems that do aggregate it either apply hard eligibility cutoffs or classify the student in isolation, without comparing the student's actual profile against what a scholarship's own description states it values. No system reviewed in Section 2 tells a student why a particular scholarship was or was not recommended in terms the student can act on.

Formally, given a student profile s and a registry of scholarships $D = \{d_1, d_2, \dots, d_n\}$, where each d_i includes both structured eligibility fields and a free-text description, the system must (i) compute a semantic-similarity score between s and each d_i using a learned text-embedding function, (ii) combine this score with structured features into a joint feature vector x_i , (iii) estimate a success probability $P(\text{success} \mid x_i)$ for each scholarship using an ensemble of predictive models, (iv) return the top- k scholarships ranked by this probability, and (v) for every returned recommendation, produce a ranked attribution of which features in x_i contributed most to the predicted probability all computed at a latency suitable for interactive use.

4 Proposed System Architecture

Two design choices in this pipeline are not evident from the module list alone. First, the eligibility gate operates **per student–scholarship pair**, not per student, since a student may be eligible for one scholarship but ineligible for another due to scholarship-specific criteria. Second, the gate is placed **before semantic matching and ensemble prediction**, which are the two most computationally expensive stages. Filtering structurally ineligible pairs first avoids unnecessary SBERT encoding and model inference, keeping per-recommendation latency dependent mainly on the number of eligible candidates rather than the full registry size, as required by Section 3.

Ineligible candidates are not discarded but follow a narrower path through the same architecture. The **Gap Analysis and Roadmap** module reuses the structured comparisons from the Eligibility Engine, converting failed criteria into signed gaps (e.g., required GPA minus actual GPA) and mapping them to preparatory or bridging courses through the Program Catalog. Thus, both branches terminate at **Personalized Output**: the eligible branch provides a ranked and explained shortlist of currently applicable scholarships, while the ineligible branch provides a diagnostic roadmap for

future eligibility. Relative to Section 3, the eligible branch implements steps (i)–(v), while the ineligible branch replaces steps (i)–(iii) with gap-vector computation and steps (iv)–(v) with roadmap generation. This shared structure allows the six core modules Student Profile, Feature Processing, Semantic Matching, Prediction, Recommendation, and Explainable AI together with the eligibility gate, to provide personalized and actionable outputs for every candidate rather than only an unexplained pass/fail result. This is explained in the below Fig 2

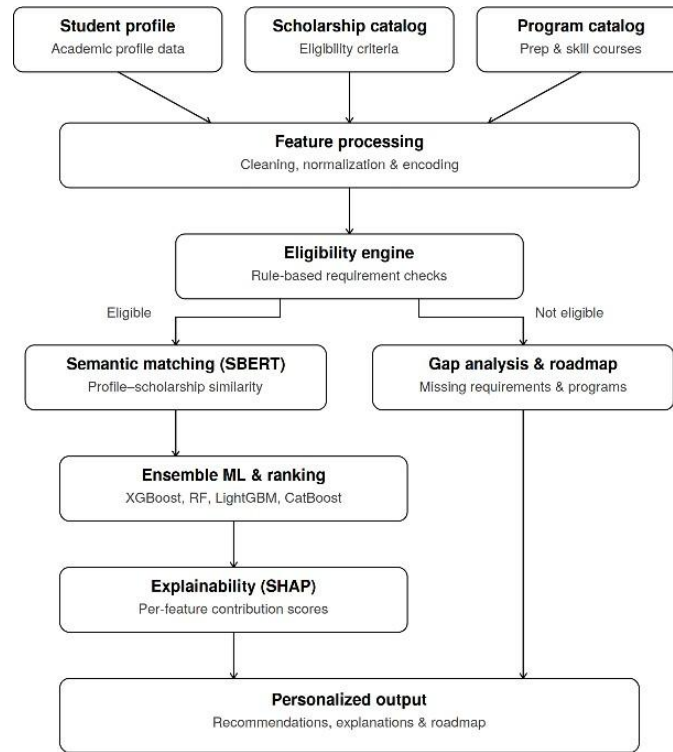


Fig. 2. Proposed system architecture and workflow.

4.1 Student Profile Module

The pipeline begins with the Student Profile module, which collects five categories of raw input: academic details (grades, standardized test scores, credits completed), personal information, extracurricular and achievement records, financial background, and, where applicable, stated preferences and goals. This module is intentionally broad, capturing both the quantitative fields a Boolean filter would use and the descriptive, free-text fields essays, activity descriptions, personal statements on which the semantic-matching module later draws.

4.2 Feature Processing Module

Raw profile data undergoes data cleaning to remove malformed or missing entries, text processing to prepare free-text fields for embedding, normalization to rescale numeric fields onto a common range, and encoding to convert categorical fields (admission type, category, gender) into a machine-readable form. The output of this module is a clean, structured student-feature vector ready for both the semantic-matching and prediction stages downstream.

4.3 Scholarship-Specific Matching Criteria

Each scholarship is associated with distinct eligibility and preference criteria, allowing the proposed system to evaluate a student's suitability for individual scholarships rather than applying a uniform matching rule. These criteria consider academic performance, extracurricular achievements, credit completion, failed-course status, student year, and major compatibility. For example, technology- and STEM-oriented scholarships assign greater importance to academic performance and relevant majors, whereas leadership- and community-oriented scholarships emphasize extracurricular participation and leadership-related achievements.

The common eligibility requirements for all scholarships are as follows:

- $\text{GPA} \geq 7.0$
- $\text{Total credits} \geq 14$
- No failed courses
- $\text{Student year} \leq 4$

The system first verifies these common eligibility conditions and then evaluates scholarship-specific attributes to determine the degree of match. These criteria are subsequently combined with the semantic similarity between the student's profile and the scholarship description to generate the features used by the prediction and ranking modules.

4.4 Semantic Matching Module (SBERT)

This module most clearly distinguishes the proposed framework from the baseline. Each scholarship's free-text description is encoded into a dense embedding using Sentence-BERT [9], and the student's own free-text profile fields are embedded in the same way. A semantic-similarity score between the two embeddings, formalized in Eq. (1) in Section 5.3, is computed and treated as a first-class predictive feature, allowing the system to recognize a strong contextual fit for instance, between a scholarship description that emphasizes demonstrated community leadership and a student profile describing exactly that kind of activity even when no single structured field captures it directly.

4.5 Prediction Module (Four-Model Soft-Voting Ensemble)

The student-scholarship feature vector, eleven dimensions wide once the semantic-similarity score is combined with the structured attributes, is passed to four independently trained models XGBoost [10], Random Forest [11], LightGBM [12], and CatBoost [13] each of which outputs its own estimate of the probability that the student will be awarded that scholarship. A soft-voting scheme averages these four probability estimates into a single predicted success probability per scholarship, formalized in Eq. (2) in Section 5.4. This design reduces the risk that any single model's particular biases or blind spots dominate the final prediction.

4.6 Recommendation Module (Ranking)

Having obtained a predicted success probability for each candidate scholarship in the registry, the Recommendation module ranks the candidates in descending order and returns a Top-K shortlist to the student, where K is a hyperparameter selected by the student at runtime (e.g., Top-1, Top-3, or Top-5). This Top-K shortlisting represents a significant improvement over the baseline's binary-qualification approach, as it provides the student with nuanced options ordered by estimated success probability.

4.7 Explainable AI Module (SHAP)

For every scholarship returned in the Top-K shortlist, the Explainable AI module applies SHAP (SHapley Additive exPlanations) to the ensemble's prediction, decomposing the predicted success probability, per Eq. (3) in Section 5.5, into the individual contribution of each of the eleven input features. The output is a ranked, signed list of feature contributions for example, a strong GPA and high semantic similarity pushing the probability up, while a lower extracurricular score pulls it down slightly that directly answers the question a student is most likely to ask: why was this scholarship recommended to me?

5 Mathematical Formulation and Implementation Details

5.1 Dataset Description and Preprocessing

The dataset combines student academic records with a registry of ten scholarship programs. Each student is paired with all ten scholarships, producing 92,900 student-scholarship interactions from 9,290 unique students, as summarized in Table 2.

Table 2. Dataset Summary Statistics

Attribute	Value
Student records	78,039
Unique students	9,290
Scholarships	10

Student–scholarship pairs	92,900
Features	11
Target/label	application_outcome (binary)
Not awarded	74,500 (80.19%)
Awarded	18,400 (19.81%)
Semantic-similarity range (SBERT)	0.2596–0.5855
Train/Test split	70% / 30%

The eleven features comprise GPA, extracurricular points, total credits, failed-course status, student year, semester, encoded class, academic weight, extracurricular weight, major match, and SBERT semantic similarity. The application_outcome label is binary and was synthetically generated using a composite fit score, a logistic transformation, and Gaussian noise, owing to the absence of real historical award outcomes. SBERT model all-MiniLM-L6-v2 was used for semantic similarity, and class weighting was applied to address class imbalance. The data were split at the student level using random_state = 42 to prevent data leakage.

5.2 Feature Vector Construction

Each student-scholarship pair is represented as an eleven-dimensional feature vector combining structured attributes (GPA, extracurricular points, total credits, failed-course status, student year, semester, encoded class, academic weight, extracurricular weight, and major match) with the SBERT-derived semantic-similarity score described in Section 4.4. Table 4 in Section 7.4 reports the relative contribution of five of these features GPA, semantic similarity, major match, credits completed, and extracurricular score for a representative recommendation, illustrating how structured and semantic signals combine in the final prediction.

5.3 Semantic Similarity Formulation

Let $e(\cdot)$ denote the SBERT sentence-embedding function, which encodes a given text string into a fixed-dimensional dense vector. For a student-profile text t_s and a scholarship-description text t_d , the semantic-similarity feature is computed as the cosine similarity between the two embeddings:

$$\text{sim}(t_s, t_d) = [e(t_s) \cdot e(t_d)] / [\|e(t_s)\| \cdot \|e(t_d)\|] \quad (1)$$

where $e(t_s) \cdot e(t_d)$ denotes the inner product of the embedded vectors and $\|\cdot\|$ denotes the L2-norm operator. Because SBERT is a contextualized embedding model, this computation captures the degree to which two texts are semantically similar: texts that are lexically different but semantically related still yield a high cosine-similarity score.

5.4 Soft-Voting Ensemble Formulation

For a given student-scholarship pair with feature vector x , each of the four base learners M_j , for $j \in \{\text{XGBoost, Random Forest, LightGBM, CatBoost}\}$, outputs an estimated probability of success $P_j(\text{success} \mid x)$. The soft-voting ensemble then computes an aggregated predicted probability as an unweighted average:

$$P(\text{success} \mid x) = (1/4) \sum_{j=1}^4 P_j(\text{success} \mid x) \quad (2)$$

This averaged probability is what the Recommendation module (Section 4.6) uses to rank scholarships. Soft voting was chosen over hard (majority-label) voting because it preserves each base learner's confidence rather than collapsing it to a single class, which matters directly for ranking: two scholarships that a model would both label as “qualifying” can still be meaningfully ordered by the ensemble's relative confidence in each.

5.5 Explainability Formulation

Given the ensemble's prediction function P for a specific student-scholarship pair x , SHAP attribution decomposes the prediction into an additive sum of per-feature contributions ϕ_i , such that

$$P(x) = \phi_0 + \sum_i \phi_i \quad (3)$$

where ϕ_0 is the model's base value and each ϕ_i quantifies that feature's marginal contribution to the deviation from the base value under the Shapley-value game-theoretic formulation. Because the prediction function is itself an average over four base learners, ϕ_i is computed as the corresponding average of each base learner's individual SHAP attribution for that feature, which keeps the explanation consistent with the ensembled probability the student actually sees.

6 Technology Stack

The student-facing application is a responsive web client that surfaces the ranked scholarship shortlist together with each recommendation's SHAP-based explanation. The semantic-matching module is implemented using a pretrained Sentence-BERT model from the sentence-transformers library, while the four base learners in the prediction ensemble are implemented using XGBoost, scikit-learn's Random Forest, LightGBM, and CatBoost, respectively, combined through a soft-voting wrapper. SHAP is applied post hoc to each base learner's output and averaged as described in Section 5.5. Structured student and scholarship records are persisted in a relational data store, while scholarship descriptions and their precomputed SBERT embeddings are indexed separately to keep semantic-similarity lookups fast as the registry grows. Trained model artifacts, including the four base learners and their ensemble weights, are versioned to support retraining as new outcome data becomes available.

7 Results and Discussion

7.1 Overall Model Performance

The performance of the individual classifiers and the proposed ensemble model was evaluated using accuracy, balanced accuracy, precision, recall, F1-score, ROC-AUC, and average precision. The results are presented in Table 3.

Table 3. Overall Performance of Individual Classifiers and the Proposed Ensemble Model

Model	Accuracy	Balanced Acc.	Precision	Recall	F1	ROC-AUC	Avg. Precision
XGBoost	0.8774	0.8900	0.6321	0.9110	0.7463	0.9211	0.7552
Random Forest	0.8780	0.8894	0.6341	0.9083	0.7468	0.9212	0.7518
LightGBM	0.8764	0.8894	0.6299	0.9109	0.7448	0.9212	0.7558
CatBoost	0.8755	0.8905	0.6273	0.9154	0.7444	0.9225	0.7582
Ensemble	0.8773	0.8909	0.6314	0.9134	0.7466	0.9217	0.7571

Fig. 3 presents the comparative performance of XGBoost, Random Forest, LightGBM, CatBoost, and the proposed ensemble in terms of accuracy and balanced accuracy. The proposed four-model soft-voting ensemble achieves an accuracy of 87.73% and a balanced accuracy of 89.09% on the evaluation set, demonstrating consistent classification performance across the evaluated classes. The relatively higher balanced accuracy compared with accuracy further indicates that the proposed framework provides improved class-balanced prediction performance.

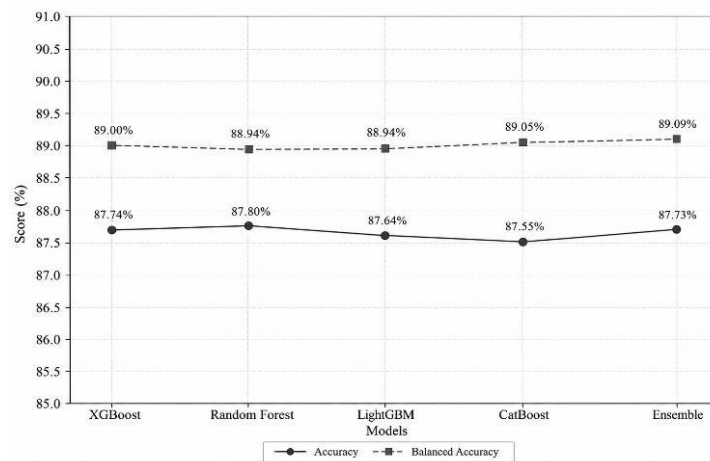


Fig. 3. Comparison of accuracy and balanced accuracy across the evaluated models.

7.2 Precision, Recall, and F1-Score Analysis

Fig. 4 compares the precision, recall, and F1-score of the five evaluated models. The high recall indicates that the ensemble is effective at identifying positive scholarship outcomes, while the F1-score provides a balanced assessment between precision and recall. Although Random Forest achieves a marginally higher precision of 63.41% and F1-score of 74.68%, the proposed ensemble obtains a higher recall of 91.34%, demonstrating its effectiveness in reducing missed positive recommendations.

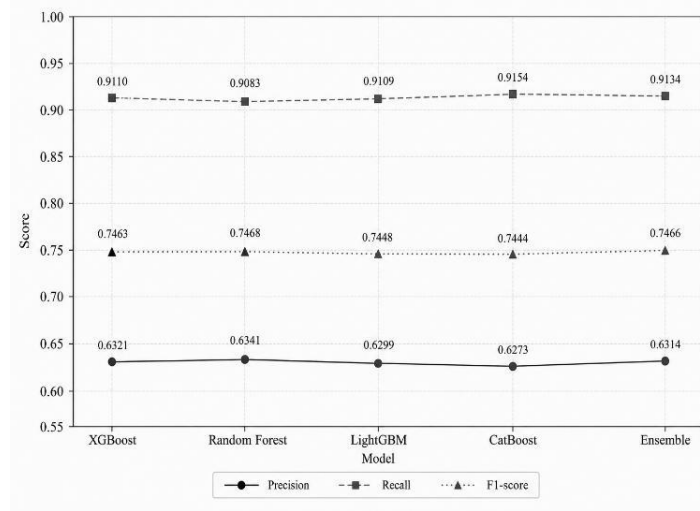


Fig. 4. Comparison of precision, recall, and F1-score across the evaluated models.

7.3 Baseline and Proposed System Comparison

To assess the improvement achieved by the proposed framework, its balanced accuracy was compared with that of the baseline XGBoost-based system. The baseline system achieves a balanced accuracy of 86.75%, whereas the proposed ensemble framework achieves 89.09% an improvement of 2.34 percentage points. This improvement demonstrates the benefit of combining multiple learners with enhanced feature representations within the proposed scholarship-recommendation framework, as illustrated in Fig. 5.

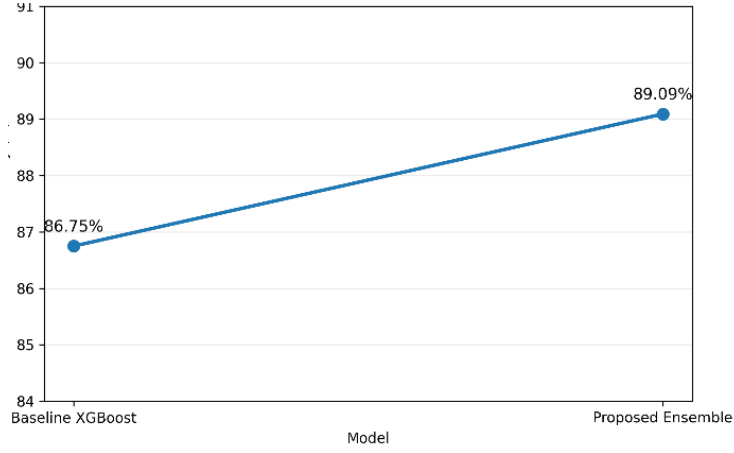


Fig. 5. Comparison of balanced accuracy between the baseline XGBoost model and the proposed ensemble learning framework.

7.4 Explainability Analysis Using SHAP

The proposed framework incorporates SHAP (SHapley Additive exPlanations) to improve the interpretability of scholarship recommendations. SHAP identifies the contribution of individual features to the model's prediction, providing insight into the factors that influence scholarship suitability. Table 4 reports the SHAP contribution values for one example recommendation, illustrating how structured academic signals and the SBERT-derived semantic-similarity score jointly drive the predicted success probability.

Table 4. Representative SHAP Feature Contributions for an Example Recommendation

Feature	SHAP Contribution
Semantic similarity (SBERT)	+0.28
GPA	+0.32
Major match	+0.18
Credits completed	+0.12
Extracurricular score	+0.10

In this example, GPA and the semantic-similarity score are the two strongest positive contributors, together accounting for more than half of the total positive contribution to the predicted probability, followed by major match, credits completed, and extracurricular score. A higher positive contribution indicates that the corresponding feature more strongly influenced the recommendation; presenting this breakdown directly to the student is what distinguishes the proposed framework's output from the baseline's unexplained qualification label. This pattern also confirms that the semantic-similarity feature introduced in Section 4.4 is not a marginal addition to the model: it

ranks as the second most influential feature in this example, ahead of three structured attributes on which the baseline system would have relied exclusively.

Taken together, these results support the central argument of this paper: reframing scholarship recommendation as a semantic-matching-plus-ensemble problem, rather than a single-classifier qualification task, improves both predictive accuracy and the transparency of the resulting recommendation, without requiring the student to interpret an unexplained label.

7.5 Top-K Ranking Performance

The ranking performance of the proposed ensemble was evaluated using Precision@K, Recall@K, and Normalized Discounted Cumulative Gain (NDCG@K) for $K = 1, 3$, and 5 , reflecting the system's actual behavior of ranking scholarships by eligibility and semantic fit before returning a Top-K shortlist. Precision@K decreases slightly from 0.2415 at $K = 1$ to 0.2241 at $K = 5$, while Recall@K rises substantially from 0.0563 to 0.2521 over the same range, and NDCG@K improves from 0.2415 to 0.2671 . This pattern is consistent with the expected precision–recall trade-off of ranking systems: as the shortlist grows from one scholarship to five, the proportion of relevant scholarships retrieved increases considerably, while the ranking quality, captured by NDCG, continues to improve rather than degrade. Together, these results indicate that the ensemble is not merely separating “qualified” from “unqualified” scholarships but is meaningfully ordering candidates by relevance, retrieving a substantially larger share of relevant scholarships as K increases without a corresponding loss in ranking quality.

Table 5. Top-K Ranking Performance of the Proposed Ensemble

K	Precision@K	Recall@K	NDCG@K
1	0.2415	0.0563	0.2415
3	0.2288	0.1541	0.2505
5	0.2241	0.2521	0.2671

8 Limitations and Challenges

Three limitations shape the current design and are the focus of ongoing work. First, the SBERT embedding step adds computational overhead relative to the baseline's structured-feature-only pipeline, particularly as the scholarship registry grows; caching precomputed scholarship embeddings mitigates most of this cost, but embedding freshly added scholarships and re-ranking against a large registry in real time remains an open engineering challenge. Second, averaging SHAP attributions across four structurally different base learners, as described in Section 5.5, is a reasonable approximation of the ensemble's reasoning but is not a perfectly faithful decomposition of a genuinely non-linear soft-voting prediction; more rigorous ensemble-level expla-

nation techniques are an area for future refinement. Third, the reported 87.73% accuracy reflects performance on the evaluation set available at this stage of the project, and broader validation against a larger, more demographically diverse applicant pool is needed before the result can be generalized with confidence. Fourth, the dataset uses synthetically generated labels due to the unavailability of real historical scholarship award outcomes.

9 Conclusion and Future Scope

This paper presented an Explainable Ensemble Learning Framework for Scholarship Recommendation and Success Prediction that replaces a single-classifier baseline with a six-module pipeline: student-profile ingestion, feature processing, SBERT-based semantic matching, a four-model soft-voting ensemble, Top-K ranking, and SHAP-based explainability. Against a baseline system that predicts scholarship qualification using XGBoost alone, the proposed ensemble achieves 87.73% accuracy and a balanced accuracy of 89.09%, while additionally returning a ranked shortlist and a per-feature explanation for every recommendation capabilities the baseline architecture does not provide at all. Future work includes extending the scholarship registry with live provider integrations, validating the ensemble on a larger and more diverse applicant dataset, exploring ensemble-level (rather than per-model-averaged) explanation techniques, and reintroducing a counterfactual-guidance layer that translates a low predicted success probability into concrete, actionable steps a student could take to improve standing for a given scholarship.

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Disclosure of Interests

The authors have no competing interests to declare that are relevant to the content of this article.

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